

**SRH Hochschule Berlin**

**Quality of Life Analysis: A Statistical Report**

Ilf Ali Momin (3122116)

Taseeb Ali (3122190)

(BST-BCS-12A)

Statistics

Kristian Rother

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## Introduction

Quality of life is a crucial measure that determines the well-being, happiness, and overall satisfaction of individuals within a society. It is a broad concept that includes economic, social, and environmental factors that collectively shape the daily experiences of people across the world. Various institutions, including governments, non-governmental organizations, and economists, continuously study these factors to develop policies that enhance living conditions globally.

The primary goal of this study is to analyze the quality of life across different countries based on a variety of indicators. By using statistical data and visualization techniques, we aim to understand how purchasing power, healthcare, safety, climate, cost of living, and pollution contribute to the overall quality of life. This report provides a thorough examination of global trends, highlights significant correlations, and discusses potential solutions for improving societal well-being.

## Data Overview

The dataset used for this study consists of 236 countries, covering various factors that contribute to the overall quality of life. The following are the key indicators considered in this research:

- **Purchasing Power:** Measures the ability of individuals to buy goods and services based on their income levels. Higher purchasing power generally indicates a better economic situation.
- **Safety:** Crime rates and general security conditions determine the safety levels of a country. Nations with low crime rates tend to have a higher quality of life.
- **Healthcare:** Availability and accessibility of healthcare services, quality of hospitals, and medical infrastructure.
- **Climate:** Weather conditions significantly impact livability, with moderate and stable climates generally being preferred.
- **Cost of Living:** Assesses the affordability of essential goods and services, such as housing, food, and utilities.
- **Traffic Commute Time:** The amount of time people spend commuting daily affects work-life balance and overall happiness.
- **Pollution:** Measures air quality and environmental conditions. High pollution levels negatively affect public health and reduce life expectancy.

- **Overall Quality of Life Score:** A composite metric that takes into account all the above factors to assess the general well-being of a country's population.

By examining these factors collectively, we can identify which aspects contribute most to improving or deteriorating the quality of life.

### **Data Cleaning and Processing**

Before conducting any analysis, the dataset was cleaned and preprocessed to ensure accuracy and consistency. The following steps were taken:

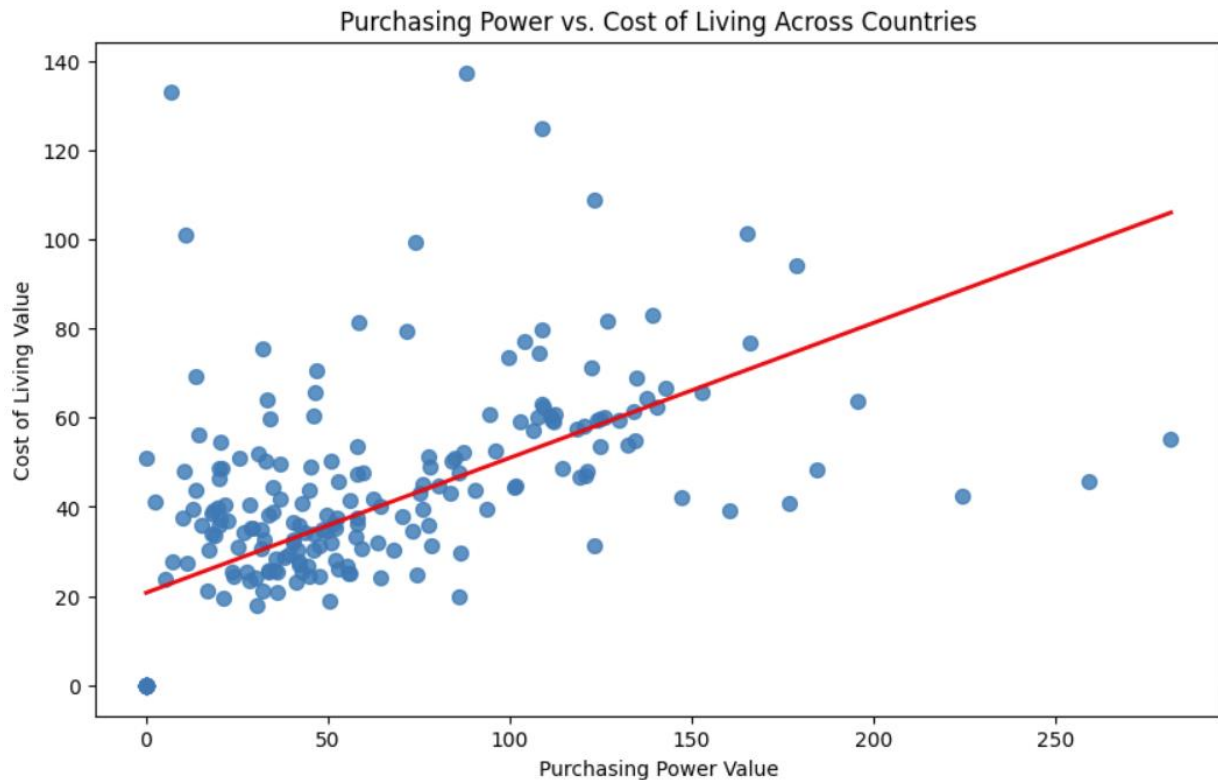
1. **Handling Missing Values:** Missing numerical values were filled using the median to prevent skewing results. Categorical variables were filled using the most frequent value (mode).
2. **Standardizing Data Formats:** Different formatting styles were corrected for uniformity.
3. **Removing Outliers:** Extreme values that could distort the analysis were identified and removed using statistical techniques.
4. **Converting Data Types:** Certain categorical variables were transformed into numerical values for better statistical interpretation.

These steps ensured that our analysis was based on a reliable and standardized dataset, minimizing errors and inconsistencies.

## Key Statistical Findings

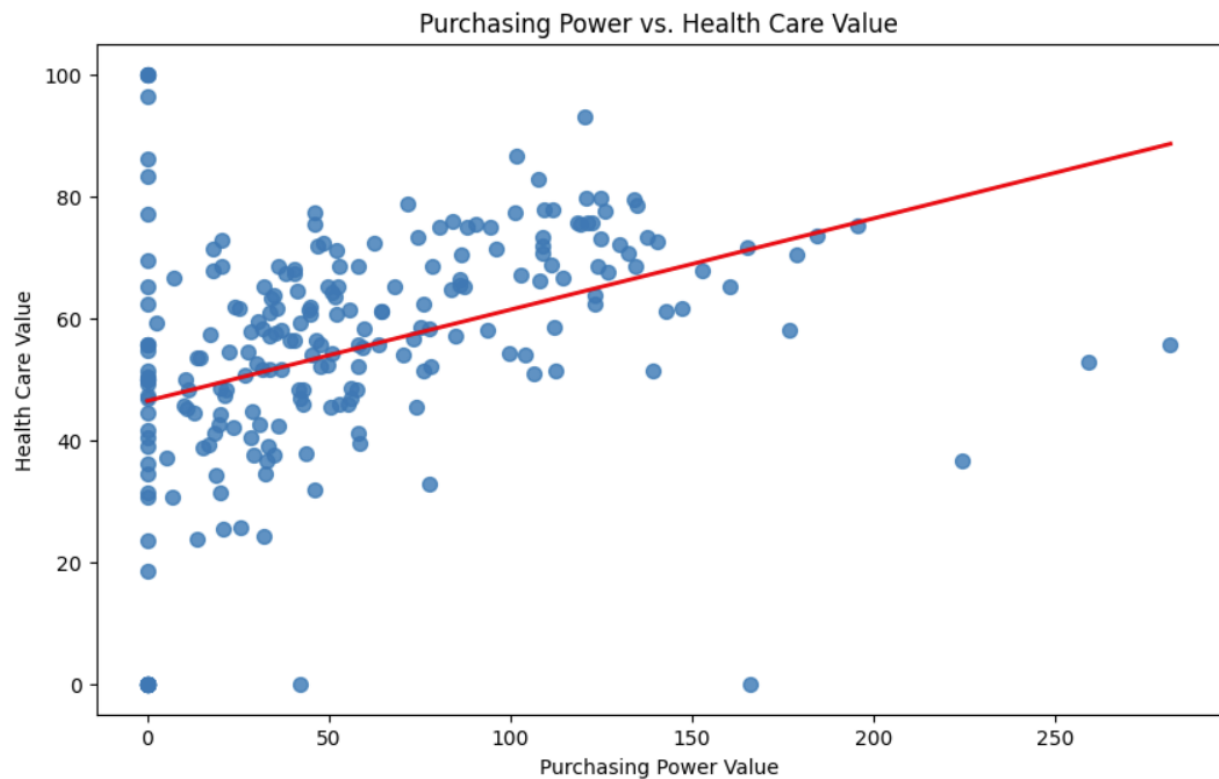
### Purchasing Power vs. Cost of Living

A regression plot was used to analyze the relationship between purchasing power and cost of living. The results show a **negative correlation**, indicating that countries with high purchasing power generally have lower relative costs of living. This means that in wealthier nations, people can afford more goods and services with their income, contributing to a higher quality of life.



## Purchasing Power vs. Healthcare

A separate regression analysis examined the link between purchasing power and healthcare quality. The data suggests that countries with higher economic power tend to have better healthcare systems due to increased investments in medical infrastructure. However, some exceptions exist where economic strength does not necessarily translate into better healthcare due to mismanagement or inefficient resource allocation.



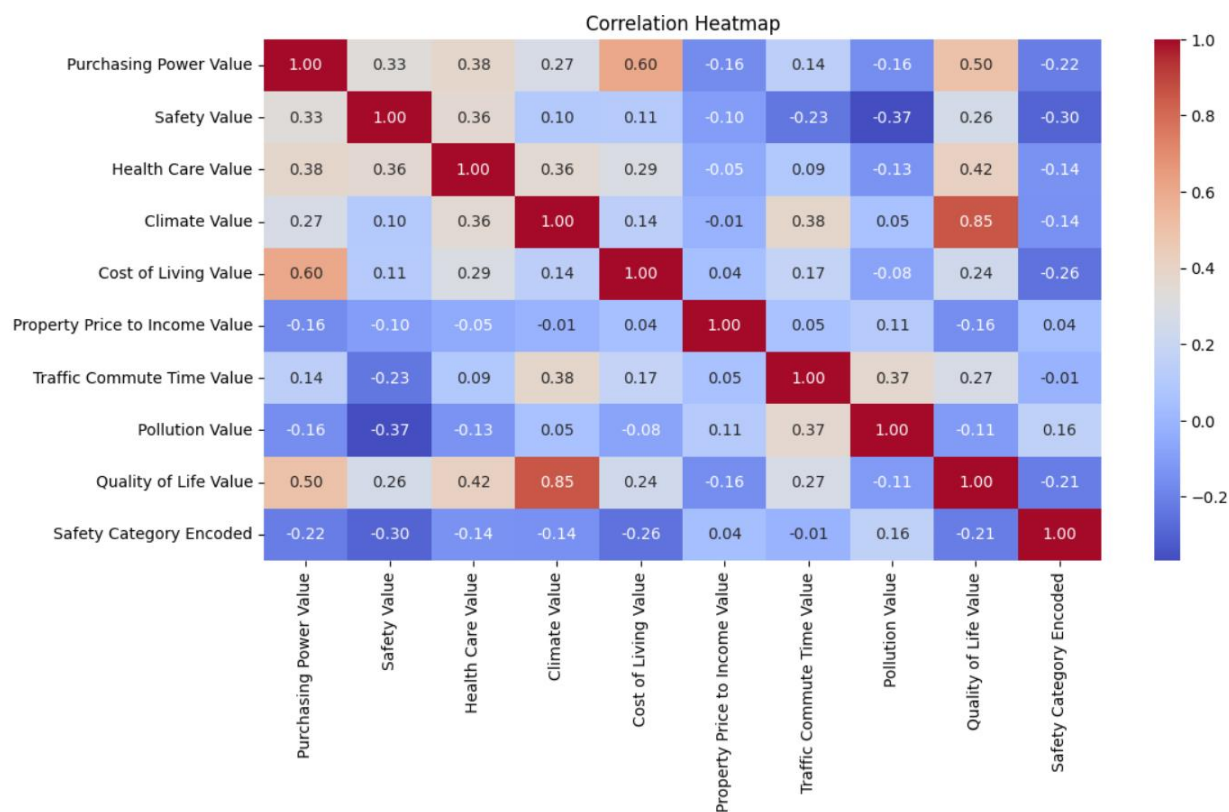
## Visual Analysis

### Correlation Heatmap

A heatmap visualization was used to highlight strong correlations between different quality-of-life indicators. Key observations include:

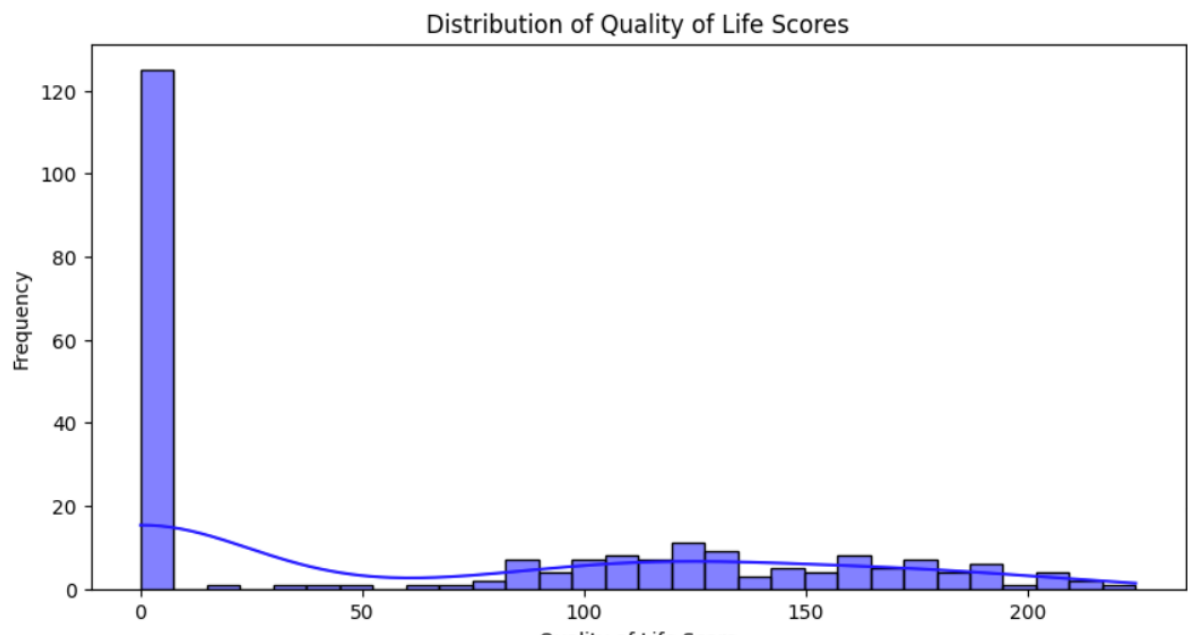
- **Pollution and traffic commute time:** Urban areas with high traffic congestion tend to have severe pollution issues.
- **Purchasing power and quality of life:** Countries with strong economies tend to have significantly higher quality of life scores.

These insights allow policymakers to identify key areas for improvement and prioritize economic and environmental policies accordingly.



**Distribution of Quality of Life Scores**

A histogram was used to analyze the distribution of quality-of-life scores across the dataset. The results reveal that some countries enjoy very high living standards, while others lag far behind. The presence of multiple peaks suggests distinct groups of nations with varying levels of well-being.



## **Distribution of Quality-of-Life Categories**

To better understand the spread of quality-of-life indicators across the dataset, we analyzed the distribution of countries across various categories for each key factor. The following observations were made:

### **1. Purchasing Power:**

- The majority of countries fall into the "Low" and "Moderate" categories, with fewer countries in the "Very High" and "Very Low" categories. This indicates that most nations have moderate economic capabilities, with a smaller number of countries at the extremes.

### **2. Climate:**

- The distribution shows that a significant number of countries have "Moderate" and "High" climate conditions, suggesting that many regions offer favorable weather for living. However, a smaller subset of countries falls into the "Very High" category, indicating extreme climate conditions.

### **3. Safety:**

- The safety distribution reveals that a large proportion of countries are categorized as "High" or "Very High" in terms of safety, reflecting relatively low crime rates and secure living conditions. However, some countries still fall into the "Low" and "Very Low" categories, highlighting areas where safety remains a concern.

### **4. Cost of Living:**

- The cost of living is relatively evenly distributed across categories, with a slight skew toward "Moderate" and "High" costs. This suggests that affordability varies significantly across different regions.

### **5. Pollution:**

- Pollution levels are distributed across all categories, with a notable number of countries in the "High" and "Very High" categories. This indicates that environmental pollution remains a significant issue in many parts of the world.

### **6. Healthcare:**

- Healthcare quality shows a diverse distribution, with many countries falling into the "Moderate" and "High" categories. However, the presence of

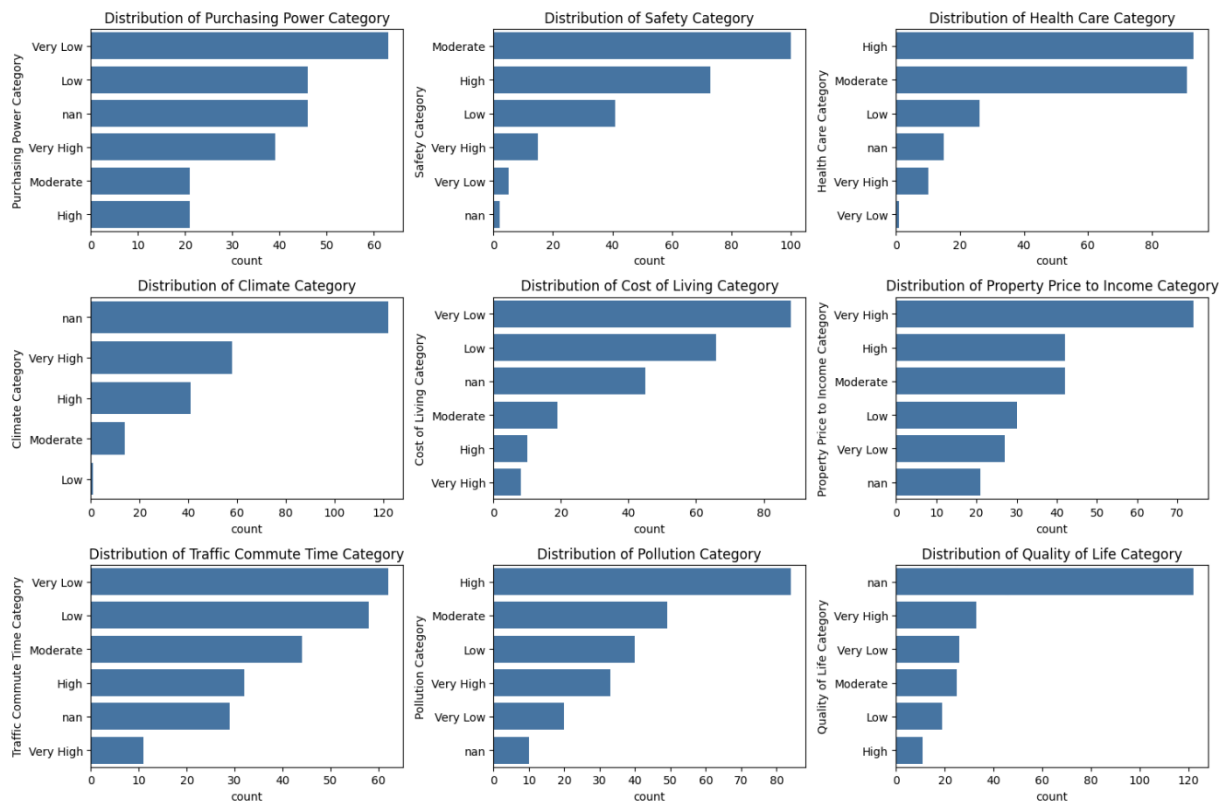


countries in the "Very Low" and "Low" categories underscores disparities in healthcare access and quality.

## 7. Quality of Life:

- The overall quality of life distribution reveals a mix of "High," "Moderate," and "Low" categories, with a few countries achieving "Very High" standards. This highlights the global disparity in living conditions and well-being.

These distributions provide valuable insights into the global landscape of quality-of-life indicators. Policymakers can use this information to identify areas where improvements are needed, such as reducing pollution, enhancing healthcare, and ensuring safety in regions that lag behind.



## Cluster Analysis and Additional Insights

To further categorize countries based on their quality of life, a **K-Means clustering analysis** was performed. The algorithm grouped countries into three main clusters:

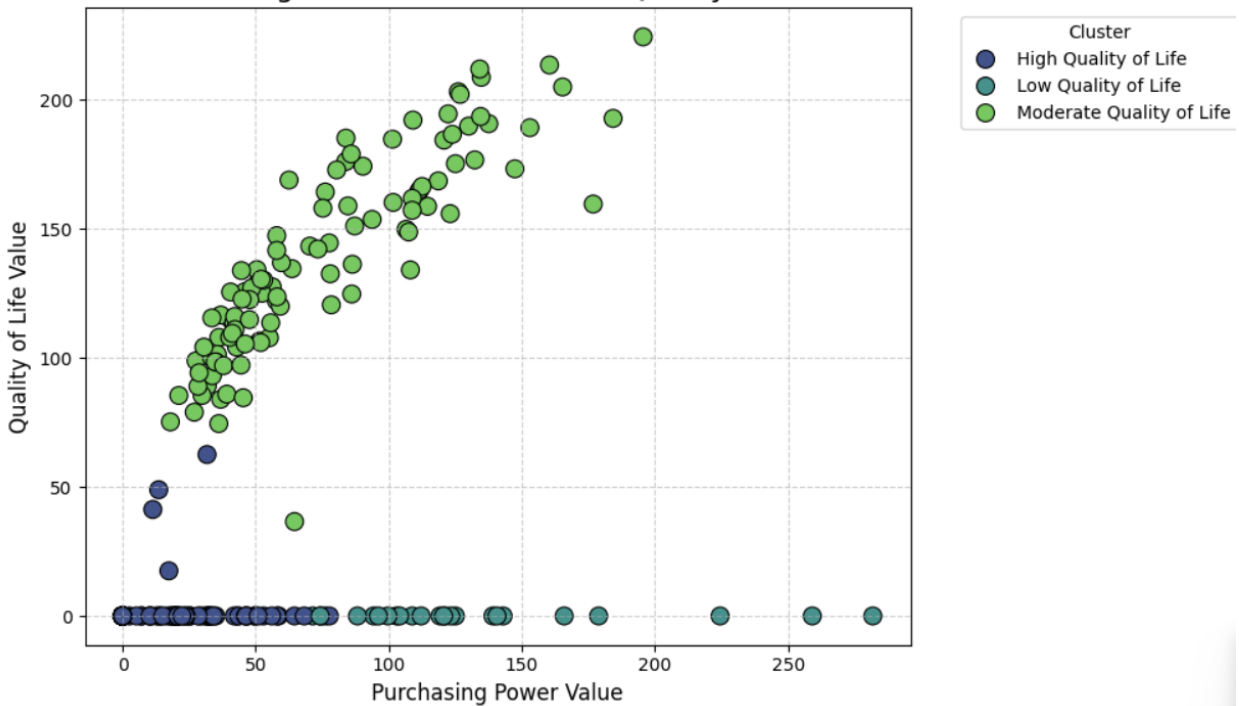
- **Cluster 1:** Countries with high purchasing power, excellent healthcare, and low pollution levels (e.g., Switzerland, Norway, Canada).

- **Cluster 2:** Countries with moderate quality of life, balanced economic conditions, and mixed pollution levels (e.g., Brazil, Mexico, Turkey).
- **Cluster 3:** Countries with low purchasing power, high pollution, and poor healthcare (e.g., India, Nigeria, Bangladesh).

These results suggest that while economic strength contributes to higher living standards, environmental factors such as pollution and commute time also play a significant role in determining quality of life.

**Recommendation:** Governments in Cluster 2 and Cluster 3 should prioritize investments in healthcare and pollution control to bridge the gap with high-quality-of-life nations.

K-Means Clustering of Countries Based on Quality of Life Indicators



## Analysis of the Geographical Distribution of Quality of Life Map

The map visually represents the Quality of life distribution across different countries, with colors indicating varying levels of quality. The color scale on the right ranges from **dark blue (lowest quality of life) to bright yellow (highest quality of life)**.

### Key Observations:

#### 1. High Quality of Life (Yellow & Orange Regions):

- Countries in North America (USA, Canada), Western Europe (Germany, France, UK, Switzerland, Norway), and Australia exhibit the highest quality of life.
- These regions typically have strong economies, excellent healthcare, and lower pollution levels, which contribute to a higher standard of living.

#### 2. Moderate Quality of Life (Pink Regions):

- Eastern Europe, parts of South America (e.g., Brazil, Argentina), and some Asian countries (China, Turkey, Russia) fall into a middle category.
- These countries have moderate economic conditions, adequate healthcare, and varying levels of pollution.

#### 3. Low Quality of Life (Dark Blue & Purple Regions):

- Many countries in Africa (Nigeria, Sudan, Democratic Republic of Congo), South Asia (India, Bangladesh, Pakistan), and parts of the Middle East show the lowest quality of life.
- These areas struggle with low purchasing power, poor healthcare infrastructure, high pollution levels, and economic instability.

### Key Insights:

- **Economic Strength & Quality of Life:**

The map suggests a strong correlation between economic power and quality of life, as wealthier nations tend to score higher.

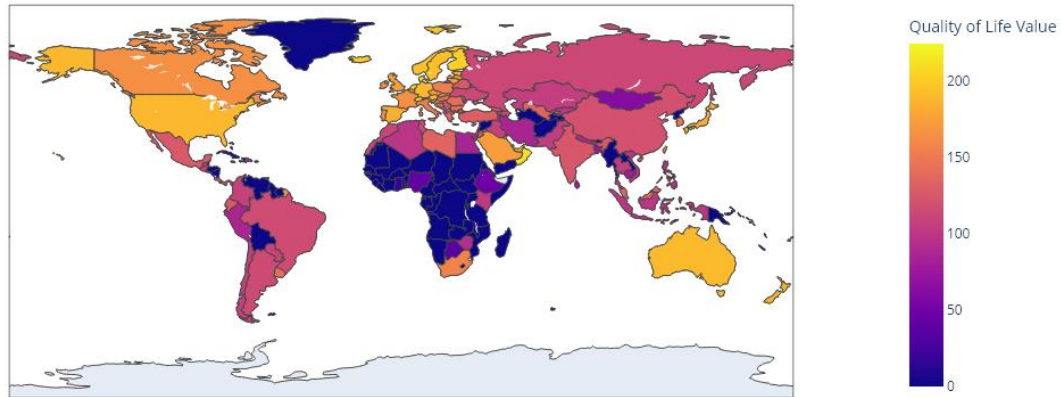
- **Environmental Impact:**

Countries with high pollution and poor infrastructure tend to rank lower in quality of life.

- **Geographical Patterns:**

- Developed nations in **North America, Western Europe, and Oceania** tend to have the highest quality of life.
- **Developing nations** in Africa, South Asia, and parts of Latin America struggle due to economic and environmental challenges.

Geographical Distribution of Quality of Life



### Future Predictions and Trends

- **Economic Growth and Well-being:** As economies continue to expand, purchasing power is expected to increase, leading to better access to healthcare and improved overall well-being.
- **Urbanization Challenges:** Many cities face issues related to traffic congestion and pollution. Without effective infrastructure planning, these problems may worsen.
- **Technological Advancements:** Innovations in healthcare, transportation, and sustainable energy will be crucial for maintaining high-quality living standards.
- **Climate Change Impact:** Climate change may negatively affect safety, health, and economic stability. Proactive measures are necessary to ensure long-term sustainability.

### Policy Recommendations

1. **Investing in Healthcare:** Governments should allocate sufficient resources to ensure equitable access to healthcare services.
2. **Enhancing Public Safety:** Strengthening law enforcement and community safety programs to reduce crime rates.

3. **Improving Urban Infrastructure:** Investing in public transportation and green spaces to reduce congestion and pollution.
4. **Tackling Environmental Issues:** Enforcing stricter pollution control regulations to protect air and water quality.
5. **Economic Stability Programs:** Policies that promote job creation, fair wages, and economic resilience will contribute to better living standards.

## **Conclusion**

This study provides an extensive analysis of quality-of-life indicators across 236 countries, highlighting key factors that impact well-being. Economic stability, healthcare access, environmental conditions, and safety are the most influential determinants of quality of life. By addressing these issues through strategic policymaking, governments can ensure a better future for their citizens.

Further research should focus on regional differences, long-term trends, and innovative solutions to continuously improve global living standards.

**Sources → <https://www.kaggle.com/datasets/ahmedmohamed2003/quality-of-life-for-each-country>**